In [1]:
```
import pandas as pd
import numpy as np
from sklearn.tree import DecisionTreeClassifier,plot_tree
from sklearn.preprocessing import LabelEncoder
import matplotlib.pyplot as plt
```

In [2]:
```
data={
    'Outlook':['Rainy','Rainy','Overcast','Sunny','Sunny','Sunny','Overcast','Rainy'
    'Temperature':['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild
    'Humidity':['High','High','High','High','Normal','Normal','Normal','High','Norma
    'Windy':['False','True','False','False','False','True','True','False','False','F
    'Play':['Yes','No','Yes','No','Yes','No','Yes','No','Yes','Yes','Yes','Yes','Yes
```

In [3]:
```
df=pd.DataFrame(data)
print(df.head())
```

```
    Outlook Temperature Humidity  Windy Play
0     Rainy         Hot     High  False  Yes
1     Rainy         Hot     High   True   No
2  Overcast         Hot     High  False  Yes
3     Sunny        Mild     High  False   No
4     Sunny        Cool   Normal  False  Yes
```

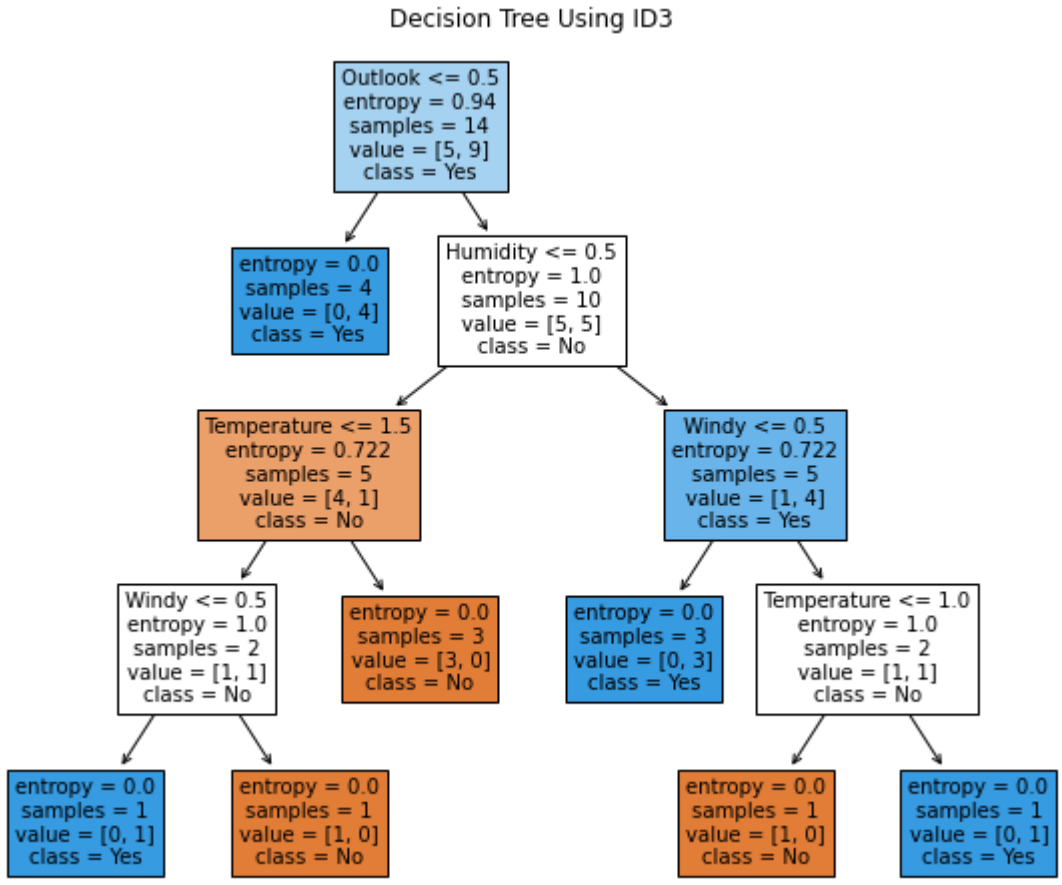
In [4]:
```
le=LabelEncoder()
for col in df.columns:
    df[col]=le.fit_transform(df[col])

x=df.drop('Play',axis=1)
y=df['Play']
```

In [5]:
```
model=DecisionTreeClassifier(criterion='entropy')
model.fit(x,y)
```

Out[5]:
```
DecisionTreeClassifier(criterion='entropy')
```

In [6]:
```
plt.figure(figsize=(10,8))
plot_tree(model,feature_names=x.columns,class_names=['No','Yes'],filled=True)
plt.title("Decision Tree Using ID3")
plt.show()
```


Decision Tree Using ID3
Outlook <= 0.5
entropy = 0.94
samples = 14
value = [5, 9]
class = Yes
entropy = 0.0
samples = 4
value = [0, 4]
class = Yes
Humidity <= 0.5
entropy = 1.0
samples = 10
value = [5, 5]
class = No
Temperature <= 1.5
entropy = 0.722
samples = 5
value = [4, 1]
class = No
Windy <= 0.5
entropy = 0.722
samples = 5
value = [1, 4]
class = Yes
Windy <= 0.5
entropy = 1.0
samples = 2
value = [1, 1]
class = No
entropy = 0.0
samples = 3
value = [3, 0]
class = No
entropy = 0.0
samples = 3
value = [0, 3]
class = Yes
Temperature <= 1.0
entropy = 1.0
samples = 2
value = [1, 1]
class = No
entropy = 0.0
samples = 1
value = [0, 1]
class = Yes
entropy = 0.0
samples = 1
value = [1, 0]
class = No
entropy = 0.0
samples = 1
value = [1, 0]
class = No
entropy = 0.0
samples = 1
value = [0, 1]
class = Yes

In [ ]: